PRD: Personal AI Helper on WhatsApp (Daily Briefing + Assistant)

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Overview

A single-user AI helper that lives in WhatsApp: it sends a short audio briefing every morning, answers any question by text or voice note, and remembers things you ask it to track.

Problem. Staying current on AI and world news takes scattered reading across many sources. General assistants live in separate apps, forget context, and don't deliver anything proactively.

Vision. One chat thread you already open daily. Mornings start with a 5–10 minute briefing (AI news first, world news second, a light segment last). You can reply "go deeper on story 2" or ask anything unrelated, and the assistant answers with context from the briefing, the web, and what it has learned about you.

Scope for v1. Built for one user (you), in English first, with a regional-language audio track as the second step. Multi-user support is a later decision.

Goals, non-goals, and success metrics

v1 succeeds if the briefing is good enough that you listen to it most mornings and the assistant is useful enough that you message it daily.

Goals

- Deliver a personalized 5–10 minute audio briefing to WhatsApp every morning.
- Answer follow-ups and general questions by text or voice note, grounded in sources with links.
- Remember what you ask it to track (reminders, watchlists, notes, recurring topics).
- Keep per-day running cost low enough to ignore (target set after the first week of real usage).

Non-goals for v1

- Live real-time voice calls (voice notes only; calls are a later experiment).
- Multi-user onboarding, billing, or a public product.

- Acting on your behalf in other apps (email, calendar writes). Read-only integrations come later.
- Financial or medical advice; the assistant gives information and sources, not recommendations.

Success metrics

Metric	Target	How measured
Briefing listened to	5 of 7 mornings	Audio played receipt / button tap
Story relevance	80% of stories rated useful	Thumbs up/down per story
Deep-dive usage	3+ follow-ups per week	Conversation logs
Faithfulness	0 unsupported claims in a weekly audit sample	Manual + LLM-judge check vs source text
Briefing ready on time	95% ready by your set time	Scheduler logs
Gate latency (Jev)	p95 under 500 ms	Request timing

User and key use cases

The user is you: an AI/ML engineer in Bangalore who wants daily AI and world context without opening ten tabs, and a helper for everyday questions and tracking.

1. **Morning briefing.** At a set time, WhatsApp shows a "briefing ready" message. Tap Play, hear 5–10 minutes: AI news, world news, then a light closing segment.
2. **Deep-dive.** During or after the briefing, say or type "go deeper on story 2" or "why does this matter for RAG?" and get a sourced, longer answer.
3. **General questions.** Ask anything ("explain this error", "compare these two frameworks", "what's the weather for my commute") and get a direct answer with links where facts are current.
4. **Tracking.** Ask it to watch a topic or keep a list ("tell me when Sarvam releases a new model", "add this paper to my reading list", "remind me Friday to review the PR").
5. **Feedback.** React to a story with a thumbs up or down, or say "less of this". The ranking adapts.

Functional requirements

Four capabilities make up v1: the briefing pipeline, the conversational agent, memory and trackers, and feedback-driven ranking.

A. Briefing pipeline (scheduled, daily)

- Ingest from configurable sources: RSS feeds, newsletters, Hacker News, arXiv, and news sites, in three sections (AI, world, fun).
- Clean text, de-duplicate near-identical stories, and store each with source, time, and full text.
- Rank with a two-stage filter: embeddings for dedupe, then Jev composite scoring for relevance, substance, and section. Take the top N per section (for example 4 AI, 3 world, 1 fun).
- Write a spoken-style script with an LLM using only the provided article text; target 140–150 words per minute for the chosen length.
- Fun segment uses sourced items (a quirky tech story, "on this day", trivia), never invented jokes or facts.
- Synthesize audio in chunks, stitch with short pauses, and keep a numbered story list so replies can refer to "story 2".
- Deliver through WhatsApp; also keep the audio and transcript in storage.

B. Conversational agent (on demand)

- Accept text and voice notes (voice notes go through speech-to-text first).
- Answer general questions with web search when facts may be current, and cite sources in the reply.
- Resolve references to the latest briefing ("story 2", "the Sarvam one") from stored episode state.
- Reply in the channel the user used: text for text, a voice note for a voice note.
- Tools: fetch story context, web search, related past stories, create and read trackers, set reminders.

C. Memory and trackers

- Store explicit user facts and preferences the user states (interests, language, briefing time, length, tone).
- Trackers: lists (reading list, to-dos), topic watches (alert when news matches), and reminders at a time.
- The user can ask "what do you know about me?", correct it, or delete anything.
- Memory writes are confirmed in chat ("Saved: briefing at 7:30").

D. Feedback and ranking

- Thumbs up/down per story via message buttons, plus free text ("less crypto").
- Feedback updates per-topic weights used in composite scoring.
- Weekly summary of what changed, so the personalization stays visible.

WhatsApp channel design

The product uses the WhatsApp Business Platform (Cloud API), directly from Meta or through a provider such as Twilio. A personal WhatsApp account isn't supported for this, and unofficial automation libraries risk a number ban. Facts below come from Meta's documentation and third-party guides read on 2026-09-30; re-check them before building.

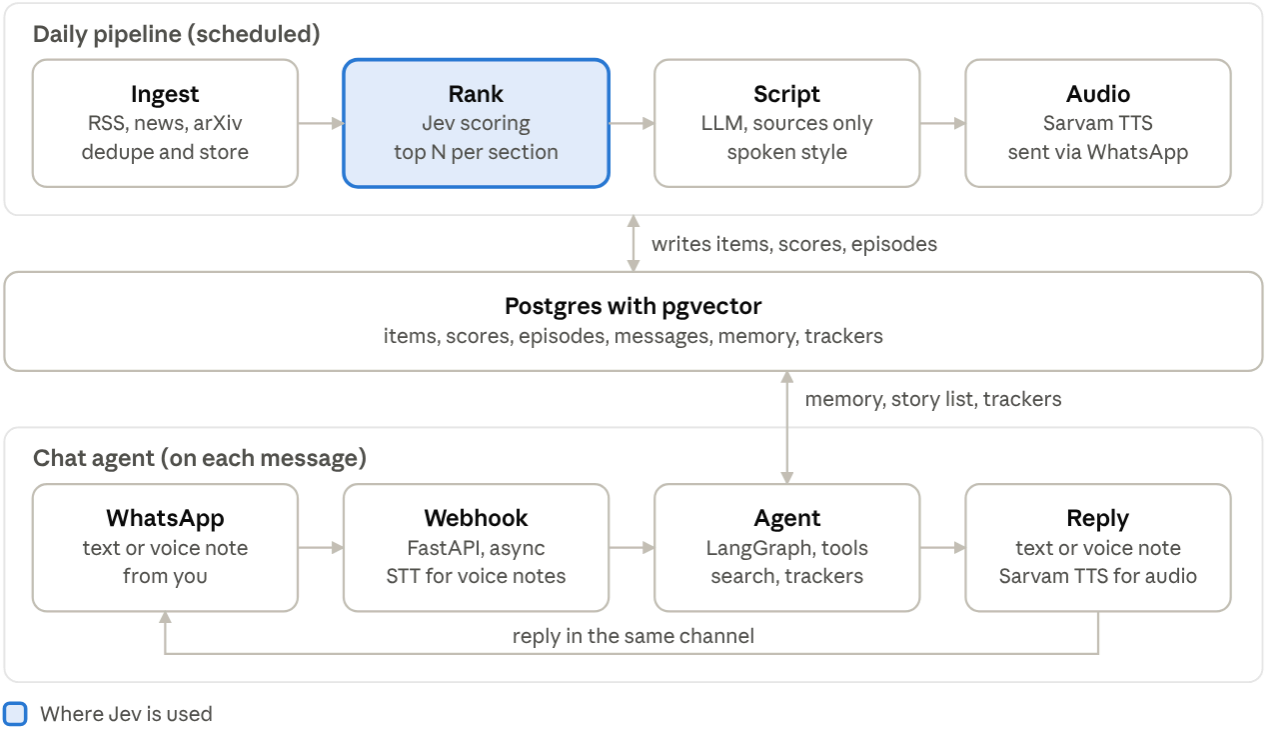
Topic	What the platform allows	Design decision
24-hour window	Free-form messages (text, audio, buttons) only within 24 hours of the user's last message; outside it, only approved template messages.	Daily push starts with a template; the user's reply or button tap opens the window.
Daily push	A template like "Your briefing is ready" with a quick-reply button (Play).	Verify in testing that a button tap opens the window as expected.
Audio delivery	Audio messages are supported; exact codec requirements were not confirmed from the docs read.	Test formats early; aim for playable voice-note display.
Voice notes in	Users can send voice notes; the webhook receives a media reference.	Download, transcribe with Sarvam STT, then run the agent.
Live calls	A Calling API exists and is available in India, but business-initiated calls need explicit user permission; AI voice-agent integration details were not confirmed.	Out of scope for v1; revisit as an experiment.
Cost	Templates are charged per message; in-window replies are cheaper or free.	Check Meta's pricing page; for one user cost should be minimal.
Development	Meta provides a test phone number for development.	Start there before business verification; confirm limits in the get-started guide.

Policy risk. Meta restricted general-purpose AI assistants on the platform in late 2025, and regulators then pushed back; the current state is reportedly open but still shifting. A personal single-purpose assistant carries lower risk, but confirm current terms with the provider and keep the agent logic channel-agnostic so it can move to Telegram or a web app if rules change.

Architecture and stack

Two flows share one Postgres database: a scheduled pipeline that builds the briefing, and a webhook-driven agent that handles your messages.

The daily pipeline and the chat agent share one database



architecture · 2 flows, 1 database

Jev ranks stories in the pipeline. The agent reads the same database, so "story 2" resolves to the right article.

Layer	Choice	Note
Backend	Python, FastAPI (async)	Webhook and scheduled jobs
Agent	LangGraph	Tools: web search, story context, trackers, reminders

Layer	Choice	Note
Ranking	Embeddings for dedupe, then Jev	Jev takes a state plus Choice, Score, and Noul questions and returns probabilities and confidence; questions run in parallel
LLM	One fast model for scripts and answers	Pick after a cost and quality test; keep it swappable
Speech	Sarvam STT and TTS	TTS model bulbul:v3 accepts up to 2,500 characters per request, so chunk scripts at sentence boundaries
Messaging	WhatsApp Cloud API	Direct from Meta or through a provider
Data	Postgres with pgvector	Also stores audio links and logs
Scheduling	Cron or a cloud scheduler	Daily run ahead of your briefing time
Tracing	Langfuse (optional)	Traces per stage for debugging and cost

Data model and memory design

Postgres holds everything; a vector column (pgvector) supports dedupe and retrieval. Memory is explicit and inspectable: nothing is stored that the user can't see or delete.

Table	Purpose	Key fields
sources	Configured feeds and sites	url, type, section, active
items	Ingested stories	source_id, title, url, published_at, full_text, embedding
item_scores	Jev outputs per item	item_id, relevance, substance, section, confidence, run_id
episodes	One briefing per day	date, script, audio_url, story_order (list of item_ids), language
messages	WhatsApp conversation log	direction, type (text/voice), text, media_ref, episode_id, timestamp
memory_facts	Facts and preferences the user stated	key, value, created_at, source_message_id
trackers	Lists, topic watches, reminders	kind, payload, due_at, status

Table	Purpose	Key fields
feedback	Story ratings and free-text preferences	item_id, rating, note, created_at

Conversation state. Each inbound message loads the latest episode's story_order, so "story 2" resolves to a specific item. The agent's working memory is the recent messages plus retrieved memory_facts relevant to the question.

Memory rules. Store only what the user states or confirms. Every write is echoed back in chat. "What do you know about me?" lists memory_facts; "forget X" deletes the row. Sensitive categories (health, finances, identity documents) are not stored.

Safety, privacy, and risks

The biggest risks are spoken hallucinations, policy changes on WhatsApp, and a single-vendor dependency. Each has a mitigation built into v1.

Risk	Why it matters	Mitigation
Hallucinated claims in audio	Spoken summaries are hard to fact-check after the fact.	Script uses only provided article text; weekly faithfulness audit; sources listed in the message.
WhatsApp policy on AI assistants	Meta restricted general-purpose AI assistants in 2025 and the rules are still shifting.	Confirm current terms with the provider; keep agent logic channel-agnostic (Telegram or web fallback).
Jev dependency and unverified claims	Accuracy and latency figures are vendor-reported; architecture is proprietary.	Build your own labeled eval set; fall back to an LLM ranker if Jev is unavailable or confidence is low.
Data leaving your control	Messages, transcripts, and memory are processed by several third-party APIs (WhatsApp, STT/TTS, LLM, Jev).	Send the minimum needed per call; review each vendor's data retention terms; no secrets or work data in the assistant.
Unwanted actions	A general assistant with tools can act on a misread message.	v1 tools are read-only or write only to your own trackers; risky tools later go through a confidence-gated confirmation step.

Risk	Why it matters	Mitigation
Prompt injection via articles	Fetched web text can contain instructions aimed at the model.	Treat article text as data; strip instructions in the script prompt; screen items with a Jev Noul question before they reach the LLM.
Sensitive topics	Health, legal, and financial questions carry higher stakes.	Give information with sources and a plain caveat, not recommendations; do not store these in memory.
Runaway cost	Daily LLM, TTS, and template charges add up silently.	Per-day budget cap, usage log per stage, alert when spend passes the cap.

Work data stays out. This is a personal tool. Don't connect it to employer systems or client data.

Evals and quality metrics

Quality is checked at three points: ranking, script faithfulness, and agent answers. Ranking doubles as the Jev-versus-LLM benchmark.

1. Ranking quality (Jev vs LLM baseline). Each day, label the candidate stories yourself with "would have wanted this" (yes/no) for two weeks. Run the same candidates through Jev composite scoring and through an LLM ranker. Compare precision at the top N, stories you would have wanted that were missed, latency per batch, and cost per 1,000 items. Report where Jev loses as well as where it wins.

2. Script faithfulness. Sample briefings weekly. For each claim in the script, check it against the source text. Use an LLM judge first, then spot-check by hand. The target is zero unsupported claims.

3. Agent answers. Keep a small fixed set of 30–50 questions across deep-dive, general, and tracker tasks. Rerun after prompt or model changes and compare answer correctness, citation presence, and correct story resolution ("story 2" maps to the right item).

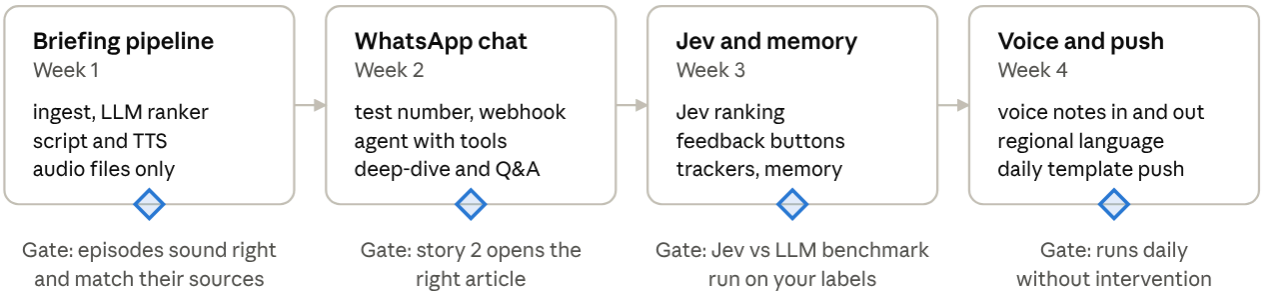
4. Confidence thresholds. For any Jev decision that triggers an action (for example, dropping an item or confirming a tracker write), plot accuracy against confidence on your labeled data, then set thresholds from that curve instead of defaults. The Jev docs suggest starting conservative (high above 0.9, medium 0.5–0.9, low below 0.5) and adjusting with your own data.

Operational metrics logged per run: stage latency, tokens and characters per provider, template messages sent, and cost per day.

Milestones and build order

Build in four phases, each ending in a gate you can check before moving on. Phase 1 uses a plain LLM ranker so Jev has a baseline to beat in phase 3.

Four phases, each ending in a gate you can check



roadmap · 4 phases, 4 gates

Week counts are rough estimates for one person building part-time; the gates matter more than the dates.

Open questions

- WhatsApp: use Meta directly or go through a provider such as Twilio?
- Data handling: are TypeSafe's terms acceptable for personal messages, and is self-hosting offered?
- Which regional language should the audio track start with?
- Single host voice, or a two-host conversation style?
- Default briefing time and length (5, 7, or 10 minutes)?
- Which LLM for scripts and answers?

Sources

Pages opened while writing this PRD (September 2026). Vendor-reported figures are unverified.

- Jev docs: Introduction
- Jev docs: Patterns
- Jev docs: Confidence
- TypeSafe: Introducing System One Models and Jev

- LangChain: Building a harness with Jev
- TechCrunch: A new kind of AI model from a ChatGPT inventor
- Sarvam: Text-to-Speech REST API
- Meta: WhatsApp service messages
- CampaignHQ: WhatsApp Business Calling API in India
- Hyyper: WhatsApp Business API 2026 updates